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17/02/2026

REG NO: 191199

DSA 8505: Bayesian Statistics

CAT 2

1. Question 1

The event of individuals on a watchlist at a busy airport.

Let

W = traveler is on the watchlist

W^c = traveler is not on the watchlist.

Historical records - 0.5% of travelers are on the watchlist.

a) Define the prior probability

$$Pr(W) = 0.5\%$$

$$Pr(W) = \frac{0.5}{100} = 0.005$$

The prior probability that a randomly selected traveler is on the watchlist is $Pr(W) = 0.005$

This is the prior probability before observing whether the system flags the traveler or not.

b) Define the likelihood of the system flagging a traveler, conditional on whether the traveler is on the watchlist or not.

F = system flags the traveler

System correctly flags a watchlist traveler with probability 95%

So likelihood if traveler is on the watchlist is

$$Pr(F|W) = 95\% = 0.95 = Pr(F|W)$$

System incorrectly flags not on the watchlist traveler with probability 3%

So likelihood is

$$Pr(F|W^c) = 3\% = 0.03 = Pr(F|W^c)$$

The likelihoods

$$Pr(F|W) = 0.95$$

$$Pr(F|W^c) = 0.03$$

c) Marginal probability that a randomly selected traveler is flagged by the system.

$$P(W) = 0.005$$

$$P(F|W) = 0.95$$

$$P(W^c) = 1 - 0.005 = 0.995$$

$$P(F|W^c) = 0.03$$

Let marginal probability be $P(F)$

$$P(F) = P(F|W)P(W) + P(F|W^c)P(W^c)$$

$$P(F) = (0.95)(0.005) + (0.03)(0.995)$$

$$= 0.00475 + 0.02985$$

$$= 0.0346$$

Approximately 3.46% of all travelers will be flagged by the system.

d) Using Bayes' Theorem, posterior probability that the traveller is actually on the watchlist given that they were flagged by the system.

$$P(W|F) = \frac{P(F|W)P(W)}{P(F)}$$

$$= \frac{(0.95)(0.005)}{0.0346}$$

$$= \frac{0.00475}{0.0346}$$

$$= 0.1373$$

$$= 13.73\%$$

Even though the system is 95% accurate for true watchlist individuals the probability that a flagged traveler is actually on the watchlist is only about 13.7%.

This is because the prior probability of 0.5% is very small.

Hence large number of non watchlist travelers leads to many false positives, reducing the posterior probability.

e) Interpret the result from a policy perspective. Discuss whether airport security should immediately detain the traveler or apply additional screening procedures.

From the posterior probability which is 13.7%, it means that only about 14 out of 100 flagged travelers are actually on the watchlist while about 86 out of 100 flagged travelers are not on the watchlist hence the false positives.

The system has a high true positive rate (95%) and a relatively low false positive rate (3%) with a prior probability of 0.5% which is extremely small.

This resulted to most flagged individuals are not truly on the watchlist. A 13.7% posterior probability means the system is helpful for screening but not strong enough evidence for immediate detention.

If the airport security detains every flagged traveler this means the majority which is about 86% would be innocent travelers.

Based on the posterior probability of 13.7%, then the airport security should not immediately detain the traveler.

Instead they should apply additional screening procedures like facial recognition system to gather more evidence before taking restrictive action.

2. Question 2.

Model 1: Linear Temperature Effect

$$M_1 = y = \beta_0 + \beta_1 x + \epsilon \quad \epsilon \sim N(0, \sigma^2)$$

Model 2: Quadratic Temperature Effect

$$M_2 = y = \beta_0 + \beta_1 x + \beta_2 x^2 + \epsilon \quad \epsilon \sim N(0, \sigma^2)$$

Equal prior probabilities for the two competing models

$$P(M_1) = P(M_2) = 0.5$$

Prior distributions for parameters

$$\beta_j \sim N(0, 10^2), \quad j = 0, 1, 2$$

$$\sigma^2 \sim \text{Inverse-Gamma}(1, 1)$$

a) Using OLS estimate regression coefficients $\hat{\beta}_0$ & $\hat{\beta}_1$ for Model M_1

where $n = 5$

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$\bar{x} = \frac{10 + 15 + 20 + 25 + 30}{5} = \frac{100}{5} = 20$$

$$\bar{y} = \frac{220 + 240 + 280 + 350 + 460}{5} = \frac{1550}{5} = 310$$

$$= \frac{(10-20)(220-310) + (15-20)(240-310) + (20-20)(280-310) + (25-20)(350-310) + (30-20)(460-310)}{(10-20)^2 + (15-20)^2 + (20-20)^2 + (25-20)^2 + (30-20)^2}$$

$$= \frac{(25-20)(350-310) + (30-20)(460-310)}{(30-20)^2}$$

$$= \frac{900 + 350 + 0 + 200 + 1500}{100 + 25 + 0 + 25 + 100} = \frac{2950}{250}$$

$$\hat{\beta}_1 = 11.8$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\hat{\beta}_0 = 310 - 11.8 \times 20$$

$$= 310 - 236$$

$$\hat{\beta}_0 = 74$$

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$$\hat{y} = 74 + 11.8x$$

For M_1 , electricity consumption increases by approximately 11.8 kWh for each 1°C increase in temperature.

b) Likelihood function for Model M_1

$$M_1: y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2)$$

Since the error term is normally distributed

$\epsilon_i \sim N(0, \sigma^2)$ it follows that:

$$y_i | \beta_0, \beta_1, \sigma^2 \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$$

$$L(\beta_0, \beta_1, \sigma^2 | y, x) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right)$$

c) Using Bayes theorem, posterior distribution of the regression parameters under M_2 .

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2)$$

for Bayes theorem

$$P(\theta | D) \propto P(D | \theta) P(\theta)$$

where $\theta = (\beta_0, \beta_1, \beta_2, \sigma^2)$

$$P_{\text{prior}} = P(\theta)$$

$$\text{likelihood} = P(D | \theta)$$

Likelihood (normal errors) $\sum$

$$P(y | \beta_0, \beta_1, \beta_2, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \beta_0 - \beta_1 x_i - \beta_2 x_i^2)^2}{2\sigma^2}\right)$$

Priors: $\beta_j \sim N(0, 10^2)$, $j = 0, 1, 2$
 $\sigma^2 \sim \text{Inverse-Gamma}(1, 1)$

Posterior distribution is

$$P(\beta_0, \beta_1, \beta_2, \sigma^2 | y, x, M_2) \propto P(y | \beta_0, \beta_1, \beta_2, \sigma^2) \cdot P(\beta_0) \cdot P(\beta_1) \cdot P(\beta_2) \cdot P(\sigma^2)$$

d) Why MCMC methods are required to obtain samples from the posterior distribution.

$$P(\beta_0, \beta_1, \beta_2, \sigma^2 | y, x) \propto P(y | \beta_0, \beta_1, \beta_2, \sigma^2) \cdot P(\beta_0) \cdot P(\beta_1) \cdot P(\beta_2) \cdot P(\sigma^2)$$

MCMC is required because

The model includes a quadratic term ✓

The likelihood depends on multiple parameters jointly

The variance σ^2 has an Inverse-Gamma prior

The resulting posterior distribution ~~does~~ not simplify to a standard distribution that can be sampled from directly.

MCMC methods are required because the posterior distribution under Model M_2 does not have a closed-form solution and involves multidimensional integrals that cannot be solved analytically. It allows to approximate the posterior distribution numerically by generating samples from it.

e) Marginal likelihoods for the two models

$$P(D | M_1) = 0.015$$

$$P(D | M_2) = 0.045$$

i) Bayes Factor in favor of Model M_2 .

$$BF = \frac{P(D | M_2)}{P(D | M_1)}$$

$$= \frac{0.045}{0.015} = 3$$

③

The observed data are three times more likely under Model M_2 than under Model M_1 .

ii) Posterior model probabilities $P(M_1|D)$ and $P(M_2|D)$

Since the prior model probabilities are equal:

$$P(M_1) = P(M_2) = 0.5$$

$$P(M_1|D) = \frac{1}{1 + BF} = \frac{1}{1 + 3} = 0.25$$

②

$$P(M_2|D) = \frac{BF}{1 + BF} = \frac{3}{1 + 3} = \frac{3}{4} = 0.75$$

After observing the data, there is 75% posterior probability in favor of the quadratic model M_2 , indicating stronger support for including the temperature squared term.

iii) Interpret the results of the Bayesian model comparison.

The Bayes factor of 3 means ① that the observed data are three times more likely under the quadratic model M_2 than under the linear model M_1 .

Since the posterior probability of M_2 is 0.75, this indicates that there is 75% probability that the quadratic model is the better model, given the data and equal prior beliefs.

Model M_2 is better supported by the data than the simple linear model M_1 .

f) Posterior mean estimator of the regression coefficients.

$$\text{Model } M_1: \hat{\beta}_0 = 150, \hat{\beta}_1 = 8.5$$

$$\text{Model } M_2: \hat{\beta}_0 = 120, \hat{\beta}_1 = 6.2, \hat{\beta}_2 = 0.45$$

i) General BMA formula for regression coefficients.

Posterior distribution of a parameter θ is

$$P(\theta|D) = \sum_k P(\theta|M_k, D) P(M_k|D)$$

$$P(M_1|D) = 0.25, P(M_2|D) = 0.75$$

For Regression Coefficient

$$\hat{\beta}_{BMA} = \sum_k \hat{\beta}^{(k)} P(M_k|D)$$

where:

$\hat{\beta}^{(k)}$ = posterior mean of the coefficient under model M_k

$P(M_k|D)$ = Posterior probability of the model M_k

ii) Compute

$$\hat{\beta}_0 = (0.25)(150) + (0.75)(120) = \sum \hat{\beta}_0 P(M|D)$$

$$= 37.5 + 90$$

$$= 127.5$$

$$\hat{\beta}_1 = (0.25)(8.5) + (0.75)(6.2) = \sum \hat{\beta}_1 P(M|D)$$

$$= 2.125 + 4.65$$

$$= 6.775$$

$$\hat{\beta}_2 = \sum \hat{\beta}_2 P(M|D)$$

$$= (0.25)(0) + (0.75)(0.45)$$

$$= 0.3375$$

iii) Final BMA regression equation.

$$\hat{y}_{BMA} = 127.5 + 6.775x + 0.3375x^2$$